

SOFIA ANCHIA

Software & Automation Engineer | AI/ML Integration | Data Science, Risk & Predictive
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PROFESSIONAL SUMMARY

Computer Science + Business engineer with hands-on experience shipping production data automation platforms, ML-driven predictive systems, and embedded hardware prototypes. Combines semiconductor manufacturing experience with full-stack AI engineering and mechanical design, backed by a track record of building systems that ship, from raw data pipelines to physical hardware.

- Built a 3-tier yield analytics automation platform at Samsung Austin Semiconductor, cutting manual data entry time by 95% (~5 hours/week) and restoring 100% data completeness on a broken production pipeline.
- Architected an end-to-end geospatial ML system processing 180K+ crash records into a live, deployed risk-detection API, and a serverless AWS AI framework integrating MCP (Model Context Protocol) that kept ongoing costs under 1% of a \$50K project budget.

AREAS OF EXPERTISE

- ▶ Full-Stack & UI/UX Development
- ▶ Data Pipeline Automation
- ▶ Business & Executive Analytics
- ▶ Cloud & Serverless (AWS)
- ▶ ML/AI Systems (RAG, LLMs, XGBoost)
- ▶ Technical Consulting & Communication

EXPERIENCE

Samsung Austin Semiconductor

Engineering Intern – Signal One, Yield Analysis | Austin, TX | 05/2026 – 08/2026

Build automated data platforms converting raw semiconductor yield data into production-grade engineering reporting for both large and small-volume product performance.

- Developed an automated yield data analysis platform in R, integrating and normalizing data from S3 and in-house cloud storage into executive-ready reports, reducing manual data entry time by ~95% (~5 hours/week, ~260 hours annually).
- Diagnosed and resolved a recurring bug affecting quarterly calculations across ~20K wafers per quarter, restoring 100% data completeness for previously missing values on key business performance products, and improving overall pipeline reliability.
- Collaborated with yield engineers to scope and implement additional features, including patching and filtering missing die-level data for mass production products.
- Built a 3-tier Yield Trend dashboard, featuring a program ID-coded overview to isolate real yield shifts from test program noise and granular drill-down tiers covering bin group and bin movement views; translated the dashboard from R to Python to align with the team's stack for future maintenance.

AI Arrive

Field Session Program, Technical Consultant - MCP Development Team | Golden, CO | 05/2025 – 07/2025

Architected and led delivery of a serverless AI research automation platform from system design through client handover.

- Architected a serverless AI framework on AWS automating pharmaceutical research report generation, integrating a RAG system with bioinformatic tools via MCP (Model Context Protocol).
- Implemented production-ready microservices using Lambda, DynamoDB, and Bedrock Claude to orchestrate PubMed searches and gene network analysis.
- Projected serverless hosting costs at \$0–200/month, keeping ongoing overhead under 1% of the project's \$50K development investment; led agile development and biweekly client meetings through handover.

PROJECTS

Geospatial Crash Risk Predictive Model + UI

Advanced Data Science, Colorado School of Mines | Golden, CO | Jan 2026 – May 2026

- Built an end-to-end geospatial crash risk system with a Google Maps-style interactive interface, processing ~180K Colorado DOT crash records to train an XGBoost classifier (5-fold cross-validation, ROC-AUC) flagging high-risk road segments statewide.
- Implemented SHAP-based explainability with LLM-generated risk summaries and deployed a Flask API + Leaflet.js map for live, route-level hotspot detection.

ShieldRx – IoT Medication Adherence System

Design Engineering, Colorado School of Mines | Golden, CO | Aug 2024 – Dec 2024

- Led a 5-person team engineering a bolt-locking mechanism and segmented rotating door for secure, single-dose delivery; as a team, selected and validated PA6 carbon-fiber-reinforced nylon housing via additive manufacturing, stress-tested to 60 MPa.
- Built an embedded control system around a Raspberry Pi Pico with dual drive motors, Bluetooth connectivity, and haptic/audio dose alerts; developed a companion MIT App Inventor app for scheduling and adherence tracking.

RESEARCH

Colorado School of Mines

Graduate Research Assistant – Generative AI | Golden, CO | 05/2023 – 01/2024

- Conducted edge case specific research for published paper "The Prospective Future Uses of Generative AI within and beyond Industry Workplaces" (arXiv:2411.11166), coding and managing multiple survey phases capturing data from 12% of the student body.

EDUCATION

B.S. Computer Science + Business, Colorado School of Mines

August 2022 – May 2026

▶ **Relevant Coursework:** Human-Robotic Interaction, Algorithms, Advanced Software Engineering, Engineering Economics, Corporate Finance

TECHNICAL SKILLS

Languages & Tools: Excel/Sheets, Python, R, SQL, Java, C/C++, Kotlin, HTML/CSS, Git, Jupyter, RStudio, VS Code, LaTeX

Systems & Domain: AWS (Lambda, DynamoDB, Bedrock, S3), Agile/SCRUM, Jira, Machine Learning/Data Science, MVVM Architecture, Wireshark